FORMATIVE ASSESSMENT – 4 (FA-4)

MATHEMATICS PROJECT WORK

Topic: Angle Subtended by an Arc at Centre vs Remaining Part of Circle

PRELIMINARY INFORMATION

Name of the Student: .....

Class & Section: **Class 9 ( )**

Roll Number: .....

Name of the Lesson: **Circles**

Name of the Guide Teacher: .....

AIM

To experimentally verify that the angle subtended by an arc at the centre of a circle is double the angle subtended by it at any point on the remaining part of the circle (∠AOB = 2∠APB).

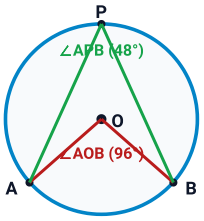
MATERIAL REQUIRED

- Drawing Sheet
- Compass & Geometry Box
- Protractor & Ruler
- Pencil & Eraser
- Pen

PROCEDURE

1. **Construction:** Draw a circle with centre O of convenient radius (e.g., 5 cm). Mark two points A and B on the circumference to define arc AB.
2. **Marking Angles:** Join OA and OB to form angle ∠AOB at the centre. Take a point P on the remaining arc and join PA and PB to form angle ∠APB.
3. **Measurement:** Measure ∠AOB and ∠APB using a protractor and calculate their ratio.

GEOMETRIC FIGURE (ANGLE AT CENTRE VS REMAINING PART)



OBSERVATION AND MEASUREMENT TABLE

S.No.	Radius of Circle	Angle at Centre (∠AOB)	Angle at Circle (∠APB)	Ratio (∠AOB / ∠APB)
1	r = 4.0 cm	**∠AOB = 80°**	**∠APB = 40°**	**80° / 40° = 2**
2	r = 5.0 cm	**∠AOB = 96°**	**∠APB = 48°**	**96° / 48° = 2**
3	r = 6.0 cm	**∠AOB = 120°**	**∠APB = 60°**	**120° / 60° = 2**

CONCLUSION

In every case, the central angle is exactly twice the angle subtended at the remaining part of the circle (∠AOB = 2 × ∠APB). Thus, the circle theorem is verified experimentally.

Signature of the Student	Signature of the Guide Teacher
Date: _____	Date: _____